

Keys and Digital Signatures

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Some administration:

- `/etc/passwd`, `/etc/shadow`, `/etc/group` and you.

Topics:

- Users
- Files
- Users, Groups, and Others (everyone else)
- basic and advanced permissions
- `sudo`, `root`, and permissions
- File Access Control Lists (ACL)

GPG last words (for now)

- Backup your `~/.gnupg` folder!
- You can export all public keys at once!
- Check your grades!
- I'm almost done exclaiming!!!
- alright, we're good now

Passwords, passwd, and you

some notes:

- `passwd` - the command to reset user passwords
- `/etc/passwd` - the file that stores user information, but NOT their password OR hashed password
- `/etc/shadow` - the file that stores user password hashes (with salts!)
- `/etc/group` - the file that stores group information including group members

Everything is a file

File Type	ID	Description
Normal	-	normal file
Directory	d	normal directory
Hard link	-	additional name for existing file
Symbolic link	l	shortcut to a file or directory
Socket	s	used to pass data between two processes
Named pipe	p	socket but users cannot work directly with pipes
Character device	c	processes character hardware communications
Block device	b	processes block hardware communications

In Linux, a user is an entity that can manipulate files (and some other things...)

Users have some of the following properties:

- Real name <- for everyone else
- User name <- for sysadmin / the user
- User ID <- (UID) what the computer actually works with
- Group(s) <- Similar to users, the computer doesn't care about the group name, it checks a GID
- password(?)

Files belong to users (and groups)

Linux file permissions are associated with the following three groups of users:

- the **U**ser that owns the file
- the **G**roup that the file belongs to
- **O**ther users (everyone else, sometimes **W**orld)
- You can change the owner and group with `chown` and `chgrp` respectively.
- You can check members of groups via the `/etc/group` file with `cat`.

Running processes have users and groups too!

This mean a process has the same permissions as the user that the process is running as!

UID's and processes

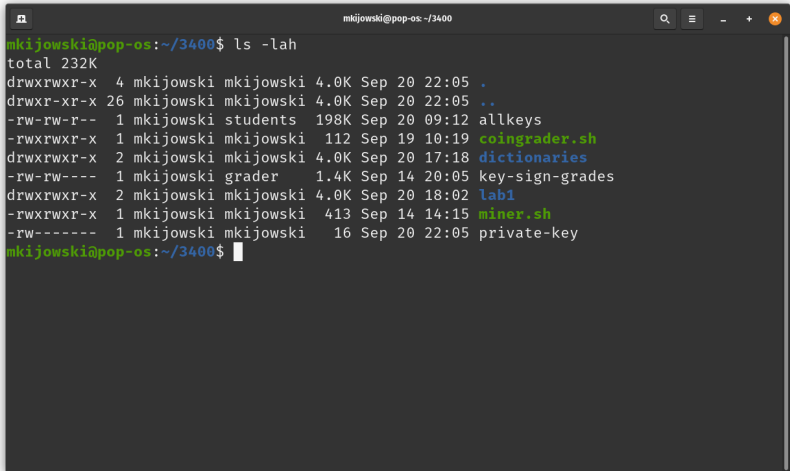
Three types of UID's

1. **Real UID:** UID of parent process (whoever is running the process)
2. **Effective UID:** Processes can gain or ched permissions by changing their UID, think passwd
3. **Saved UID:** The UID that is available to the process via some other means.

Types of permissions

3 basic types, **R**ead, **W**rite, and **eX**ecute

These are applied to each of the three groups defined previously.

A terminal window titled 'mkijowski@pop-os: ~/3400' with search, menu, and window control icons in the title bar. The terminal shows the command 'ls -lah' and its output, which lists files and directories with their permissions, owner, group, size, date, and name. The permissions are color-coded: blue for read (r), green for write (w), and red for execute (x).

```
mkijowski@pop-os:~/3400$ ls -lah
total 232K
drwxrwxr-x  4 mkijowski mkijowski 4.0K Sep 20 22:05 .
drwxr-xr-x 26 mkijowski mkijowski 4.0K Sep 20 22:05 ..
-rw-rw-r--  1 mkijowski students 198K Sep 20 09:12 allkeys
-rwxrwxr-x  1 mkijowski mkijowski 112 Sep 19 10:19 coingrader.sh
drwxrwxr-x  2 mkijowski mkijowski 4.0K Sep 20 17:18 dictionaries
-rw-rw----  1 mkijowski grader   1.4K Sep 14 20:05 key-sign-grades
drwxrwxr-x  2 mkijowski mkijowski 4.0K Sep 20 18:02 lab1
-rwxrwxr-x  1 mkijowski mkijowski 413 Sep 14 14:15 miner.sh
-rw-----  1 mkijowski mkijowski 16 Sep 20 22:05 private-key
mkijowski@pop-os:~/3400$
```

Effect of permissions on Files

Permission	Character	Description
Read	–	The file is not readable
	r	The file is readable
Write	–	The file cannot be changed or modified
	w	The file can be changed or modified
Execute	–	The file cannot be executed
	x	The file can be executed
	s	In the users triplet, setuid. If found in the group triplet,

Effect of permissions on Directories (folders)

Permission	Character	Description
Read	–	Directory contents cannot be listed
	r	Directory contents can be listed
Write	–	Directory contents cannot be altered
	w	Directory content can be altered (files can be added or removed)
Execute	–	Directory cannot be changed to (cannot cd)
	x	Directory can be

root, sudo, and security

the root user is the highest level of access on a system.

sudo or “super-user do” allows you to execute commands as the root user.

You can allow/deny access to the full sudo command in most systems with the sudo group in `/etc/group`

You can specify command specific use of sudo with the `visudo` command. This allows specified users and groups access to use all or a subset of available commands with sudo permissions.

For security reasons we want to restrict the use of root and sudo as much as possible!

- limit access to sudo to only those that need it
- limit use of sudo to only the commands that need elevated privileges

Principle of Least Privilege (PoLP)

Every program and every privileged user of the system should operate using the least amount of privilege necessary to complete the job. -Jerome Saltzer, Communications of the ACM

- Very useful principle
- Nearly impossible in practice (too much complexity)
- still useful across all areas of cyber security

- We can protect ***Confidentiality*** by restricting others access to files and limiting access to groups.
- We can protect ***Integrity*** by limiting write access to files.
- We can ensure ***Availability*** by ensuring users that need access to files are in the proper groups.

Limitations of basic Linux File Permissions

Files can only have one owner and one group.

ACL's allow for finer permissions settings per file.

setfacl and getfacl allow for additional user/group triplets.

- `setfacl -m u:<username>:<triplet> <filename>`
- `setfacl -m g:<groupname>:<triplet> <filename>`

